

Quiz 8

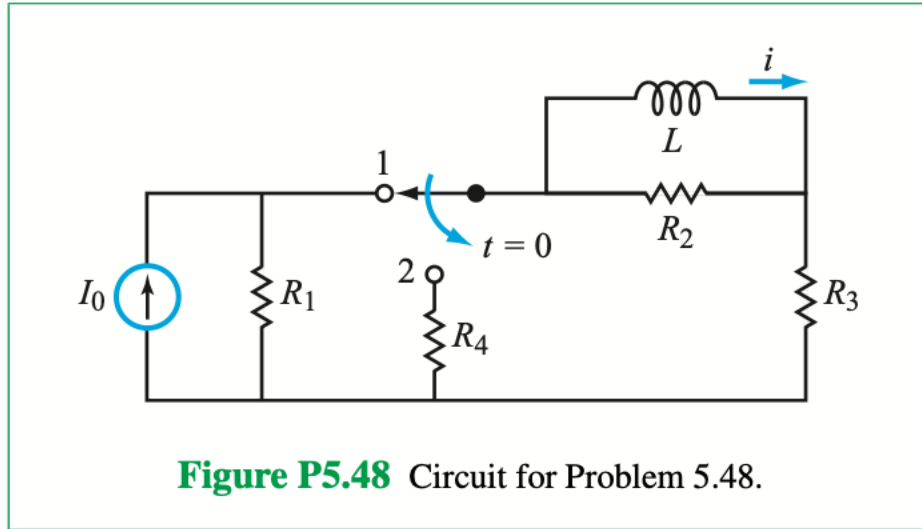
ECE 2201 – Spring 2025

Instructions

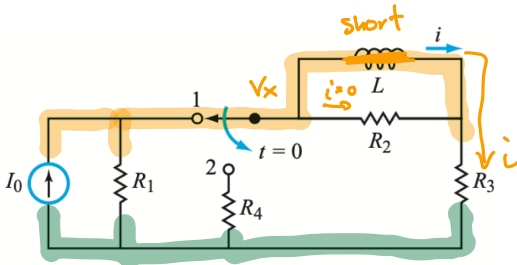
- Do not open the quiz until the instructor announces the start
- You will have 15 minutes to complete the quiz
- Closed book and notes – no formula sheets
- You may use your calculator
- Annotate your units with square brackets → for example, [A] for Amps
- Write your answer in the space provided

Name: _____ PS ID: _____

- * **5.48** Determine $i(t)$ for $t \geq 0$ given that the circuit in **Fig. P5.48** had been in steady state for a long time prior to $t = 0$. Also, $I_0 = 5 \text{ A}$, $R_1 = 2 \Omega$, $R_2 = 10 \Omega$, $R_3 = 3 \Omega$, $R_4 = 7 \Omega$, and $L = 0.15 \text{ H}$.



For $t < 0$



$$I_0 = \frac{V_x}{R_1} + \frac{V_k}{R_3}$$

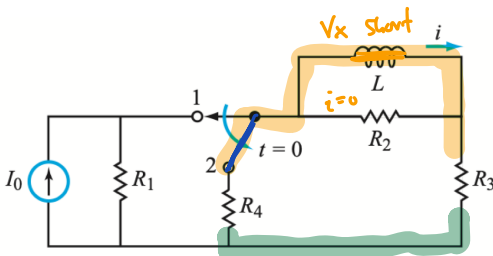
$$\left\{ 5[A] = \frac{V_x}{2[\Omega]} + \frac{V_k}{3[\Omega]} \right\} \times 6[\Omega]$$

$$30[V] = 3V_x + 2V_k$$

$$V_k = 6[V]$$

$$i'(0) = \frac{V_x}{R_3} = \frac{6[V]}{3[\Omega]} = 2[A]$$

for $t \rightarrow \infty$



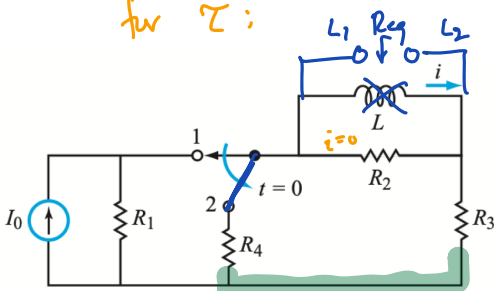
no source, so $V_k = 0$
confirm this with

$$\frac{V_k}{R_4} + \frac{V_k}{R_2} = 0$$

$$V_k = 0$$

$$i(\infty) = 0$$

for τ :



R_{eq} seen between
nodes L_1 & L_2

$$R_2 \parallel (R_3 + R_4) = 5[\Omega]$$

$$i(0) = \underline{2[A]}$$

$$i(\infty) = \underline{0}$$

$$\tau = \underline{0.03[S] = 30[ms]}$$

$$i(t) = \underline{2[A]e^{-t/30[ms]}}$$

$$\text{So, } \tau = \frac{L}{R} = \frac{0.15}{5} = 0.03[S]$$

$$= 30[ms]$$

$$i(t) = 0 + (2[A] - 0)e^{-t/30[ms]}$$